Linear Algebra

Linear Algebra

Author
MUHO CHOI
Email: sihyeon8240@gmail.com
GitHub: https://github.com/sihyeon8240

Source Code & Contributions
GitHub: https://github.com/sihyeon8240/math
Feel free to open issues or pull requests for corrections.

Contents

Preface

Throughout this textbook, **logical formulas** are interpreted according to the following conventions:

- **Parentheses** have the highest precedence. For example,

  $$(P \lor Q) \land R$$

  is evaluated by first computing $P \lor Q$.

- The **logical connectives** are interpreted in the order

  $$\neg \;>\; \land \;>\; \oplus \;>\; \lor \;>\; \Longrightarrow,\; \Longleftarrow \;>\; \Longleftrightarrow$$

  Hence

  $$P \oplus Q \implies \neg R \lor S \land T \iff U \impliedby V$$

  is understood as

  $$((P \oplus Q) \implies ((\neg R) \lor (S \land T))) \iff (U \impliedby V)$$

- A **quantifier** binds the entire formula that follows it. Thus

  $$\forall x, P(x) \land Q(x) \implies R(x)$$

  means

  $$\forall x\,((P(x) \land Q(x)) \implies R(x))$$

- The symbol $\Downarrow$ is **not** a logical connective; it is used only to indicate the flow of a proof or an explanation. For example,

  $$\begin{gathered} P \\ \Downarrow \\ Q \end{gathered}$$

  merely expresses that P is true, and therefore Q is true in the current discussion; it is not itself a proposition.

Chapter 1

Vector Spaces

1.1 Definitions

1.2 Bases

1.3 Dimension of a Vector Space

1.4 Sums and Direct Sums

Chapter 2

Matrices

Chapter 3

Linear Mappings

Chapter 4

Linear Maps and Matrices

4.1 The Linear Map Associated with a Matrix

4.2 The Matrix Associated with a Linear Map

4.3 Bases, Matrices, and Linear Maps

Chapter 5

Scalar Products and Orthogonality

5.1 Scalar Products

5.2 Orthogonal Bases, Positive Definite Case

5.3 Application to Linear Equations; the Rank

5.4 Bilinear Maps and Matrices

5.5 General Orthogonal Bases

5.6 The Dual Space and Scalar Products

5.7 Quadratic Forms

5.8 Sylvester's Theorem

Chapter 6

Determinants

Chapter 7

Symmetric, Hermitian, and Unitary Operators

7.1 Symmetric Operators

7.2 Hermitian Operators

7.3 Unitary Operators

Chapter 8

Eigenvectors and Eigenvalues

Chapter 9

Polynomials and Matrices

9.1 Polynomials

9.2 Polynomials of Matrices and Linear Maps

Chapter 10

Triangulation of Matrices and Linear Maps

10.1 Existence of Triangulation

10.2 Theorem of Hamilton-Cayley

10.3 Diagonalization of Unitary Maps

Chapter 11

Polynomials and Primary Decomposition

11.1 The Euclidean Algorithm

11.2 Greatest Common Divisor

11.3 Unique Factorization

11.4 Application to the Decomposition of a Vector Space

11.5 Schur's Lemma

11.6 The Jordan Normal Form

Chapter 12

Convex Sets

12.1 Definitions

12.2 Separating Hyperplanes

12.3 Extreme Points and Supporting Hyperplanes

12.4 The Krein-Milman Theorem

Bibliography

[1] Serge Lang, *Linear Algebra*, Springer Nature.